Jinwei Zhang

Faculty of Science and Engineering

The University of Manchester

EDUCATION

University of Manchester Manchester, UK Sep 2025 - Present

PhD in Robotics and AI: Theory of Mind and Intention Reading in Human-Robot Interaction

- **Award:** Fully funded scholarship from EPSRC Centre for Doctoral Training in Robotics and AI for Net Zero (RAINZ CDT)

University of Edinburgh Edinburgh, UK Oct 2022 - Jul 2024

MSc by Research in Informatics: IPAB: Robotics, Computer Vision, Computer Graphics and Animation

- Informatics MScR with Merit
- **Research Project:** Enhance Interpretability: A Bio-Inspired Affect-Consistent Robot Decision and Behaviour Integrated Generation Model

University of Glasgow, University of Electronic Science and Technology of China Chengdu, China

Bachelor of Engineering in Electronics and Electrical Engineering Sep 2018 - Jun 2022

- UK-China Joint Program, Math and Programming Major GPA：88 (UESTC)
  UK Upper Second Class Honours
- **Awards:** 2022 Excellent Bachelor's degree thesis of Glasgow College, UESTC, Excellent Leadership College Scholarship, Outstanding Student Scholarship of UESTC

PUBLICATION

- Jinwei Zhang, "Researches advanced in human-computer collaboration and human-machine cooperation: from variances to common prospect," in 2nd AIAHPC 2022, 10 November 2022, https://doi.org/10.1117/12.2641865
- Jinwei Zhang, J.Michael Herrmann, "A Robotic Mind Model for Affective Decision Making and Behaviour Generation", International Journal of Social Robotics 18, 23 (2026). https://doi.org/10.1007/s12369-025-01345-z

RESEARCH EXPERIENCE

Enhance Interpretability: A Bio-Inspired Affect-Consistent Robot Decision and Behaviour Integrated Generation Model Edinburgh, UK

MRes Student | Supervisor: Michael Herrmann, Lecturer | University of Edinburgh Oct 2022 – Jan 2024

- Proposed a framework for robots to react to external stimuli and have emotional waves inspired by natural creation's emotional arousal mechanism and physiology
- Designed robot motion planning and decision-making mechanism with emotional dynamics calculated
- Simulated and optimised the emotional dynamics intervened decision and behaviour integrated model in human-robot interaction scenarios based on ROS and Gazebo

Principle and Implementation of Context-Driven Human-Machine Collaboration Model Chengdu, China

Research Assistant | Advisor: Longjiang Li, Associate Professor | UESTC Sep 2021 – Jun 2022

- Designed a novel context-driven Human-Machine Collaboration model
- Optimised the Human-Machine Collaboration model by introducing human affect context into the reinforcement learning approaches of MDP and POMDP
- Simulated the collaboration model through ROS and DESPOT, a POMDP solver, and improved the simulation of human affect by adjusting the noise of observation and the degree of affect

RESEARCH COURSE PROJECTS

Robotic Systems Design Project: Autonomous Perception-Driven Exploration & Pick-and-Place

Organised by The University of Manchester MSc Robotics as RAINZ CDT training Sep 2025 – Jun 2026

- Developed an autonomous mobile robotic system capable of executing a complete "sense-plan-act" pipeline in an unknown environment, searching for and picking up colourful cubes and droping them into bins with corresponding colours
- Designed and implemented the perception subsystem using depth-camera-based computer vision to recognise target blocks, bins, bin openings, and platforms
- Developed object-attribute classification for colours and shapes, and published recognised object

coordinates obtained from the RealSense camera to the integrated ROS2 robotic framework for downstream navigation and manipulation

Intelligent tracking multiple functions rover | Leader of Patio 2
Organised by Glasgow College, UESTC and testing field provided by UESTC Mar 2021 - Jun 2021

- Wrote the main program of the tasks in Patio 2 (part two of the task field) and was responsible for the field test of Patio 2, where three tasks should be completed including matching the colour of recognized square to determine direction, releasing fish food into the lake, and communication to a laptop.
- Designed the workflow and code architecture of the second Patio tasks, led and communicated with teammates to understand the motivation functions they wrote and completed the main program utilizing the development platform STM32CubeIDE and microcontroller STM32F446RE
- Designed and assembled the fish-feeding device composed of a mechanical arm and a cup
- Tuned colour recognising and tracking functions and enhanced movement accuracy rate in task 1 from around 30% to over 90%, and calibrated compass affected by the iron handrail in the field test

LANGUAGE ABILITY AND OTHER SKILLS

- Language: English (fluent), Chinese (native)
- Skills: 8 years of programming and engineering experience, experienced in Python, C++, MATLAB, Linux system operation, ROS (Robot Operating System), ROS2, DNN